

Disproof of Conjecture 00000007709: the right triangle with legs 1 : 2 is a convex rep-5 tile

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Abstract

Conjecture 00000007709 asserts, as its first clause, that the order k of a convex rep-tile must be a perfect square or $k = 2$. This is false. The right triangle with legs in ratio 1 : 2 can be dissected into 5 congruent triangles, each similar to it with ratio $1/\sqrt{5}$, and 5 is neither a perfect square nor 2. We give the dissection with integer vertex coordinates and explicit similarities $z \mapsto (a\bar{z} + b)/5$ with $|a|^2 = 5$, and prove that it is a tiling.

A self-contained Lean 4 project (core library only) defines points, convex polygons, similarities (via squared distances) and rep- k tiles from scratch. It proves that this triangle is a rep-5 tile and derives the negation of the clause. In Lean the plane is the rational plane $\mathbb{Q}^2$; we explain why the real case follows, and the human-readable proof below is carried out directly in $\mathbb{R}^2$. We also treat the reading in which “the order” means the least $k \geq 2$. Under that reading the $1 \times \sqrt{3}$ rectangle is a counterexample: its least order is 3, and this is so even if the pieces need only be similar to the tile, not congruent to each other.

1 The conjecture

The statement (English version) reads:

Definition: The rep-tile order of a polygon is the value k for which the polygon dissects into k copies similar to itself. *Conjecture:* The order k of a convex rep-tile must be a perfect square or $k = 2$; among convex quadrilaterals the class of rep-tile order exactly 4 consists of parallelograms and three exceptional families, whose parameter spaces are two-dimensional.

The Chinese version says the same thing. The conjecture is a conjunction, so refuting one clause refutes it. We refute the first clause:

$$\text{if a convex polygon is a rep-}k \text{ tile, then } k \text{ is a perfect square or } k = 2. \quad (1)$$

We do not discuss the second clause.

2 Definitions and readings

Definition 1. A polygon $P \subset \mathbb{R}^2$ (compact, with nonempty interior) is a *rep- k tile* if there are closed sets $P_1, \dots, P_k$ such that:

1. each $P_i = f_i(P)$ for a similarity f_i of the plane;

2. the P_i are pairwise congruent;
3. $P_1 \cup \dots \cup P_k = P$;
4. the interiors of the P_i are pairwise disjoint.

Its *orders* are the k for which this holds.

This is the standard notion of Golomb [1], as used for example in [2]. As usual, “congruent” and “similar” allow mirror images. Since the areas add up, condition 2 forces every f_i to have ratio $1/\sqrt{k}$. Conversely, if every f_i has ratio $1/\sqrt{k}$, then condition 2 holds automatically, because $f_j \circ f_i^{-1}$ is an isometry mapping P_i onto P_j .

The conjecture’s wording admits several readings. We list them and say how each is handled.

- (R1) *Every order.* “The order k of a convex rep-tile” refers to any k for which the polygon is a rep- k tile. So (1) says that every order of a convex rep-tile is a square or 2. This is the reading used in the conjecture itself. Its second clause puts *all* parallelograms into “the class of rep-tile order exactly 4”. But the parallelogram with sides 1 and $\sqrt{2}$ is also a rep-2 tile, and the $1 \times \sqrt{3}$ rectangle is also a rep-3 tile. So “order 4” there means “is a rep-4 tile”, not “has least order 4”. Under (R1), Theorem 3 below refutes (1). This is the statement formalized in Lean.
- (R2) *Similar, not necessarily congruent.* The conjecture’s definition says only “ k copies similar to itself”. Dropping condition 2 gives more rep- k tiles, so a counterexample under (R1) is also one under (R2): our pieces are congruent.
- (R3) *Least order.* “The order” is the least $k \geq 2$ for which the polygon is a rep- k tile. ($k = 1$ always works, and 1 is a square anyway.) Proposition 5 gives a convex polygon whose least order is 3. This holds with congruent pieces, and also when the pieces need only be similar to the tile.
- (R4) *Orientation-preserving copies only.* The pieces in Theorem 3 are mirror images of the tile, which is allowed in the standard definition. If only orientation-preserving similarities were allowed, the rectangle of Proposition 5 still refutes the clause. Its pieces are rotated copies, and since a rectangle is mirror-symmetric the question does not arise.

The clause has the escape “ $k = 2$ ”, which does not apply to $k = 5$ or $k = 3$. There is no other exceptional set.

3 The rep-5 tiling

Identify $\mathbb{R}^2$ with $\mathbb{C}$. Let

$$A = (0, 0), \quad B = (10, 0), \quad C = (0, 5), \quad D = (2, 4), \quad M_1 = (1, 2), \quad M_2 = (6, 2), \quad M_3 = (5, 0),$$

and let $T = \triangle ABC$. This is the right triangle with right angle at A and legs $|AC| = 5$ and $|AB| = 10$, in ratio 1 : 2. Its squared side lengths are 25, 100, 125. The point D is the foot of the altitude from A : it lies on BC ($2 + 2 \cdot 4 = 10$), and $AD = (2, 4)$ is orthogonal to BC , whose direction is $(-2, 1)$. The points M_1, M_2, M_3 are the midpoints of AD, BD and AB . Define the five triangles

$$P_1 = \triangle ADC, \quad P_2 = \triangle AM_3M_1, \quad P_3 = \triangle M_3BM_2, \quad P_4 = \triangle M_1M_2D, \quad P_5 = \triangle M_3M_2M_1.$$

Geometrically, the altitude AD cuts T into P_1 (ratio $1/\sqrt{5}$) and $\triangle ABD$ (ratio $2/\sqrt{5}$). The midlines of $\triangle ABD$ cut it into four copies of ratio $1/\sqrt{5}$.

Lemma 2. For $i = 1, \dots, 5$ let $f_i(z) = (a_i \bar{z} + b_i)/5$ with

i	1	2	3	4	5
a_i	$-1 - 2i$	$2 - i$	$2 - i$	$2 - i$	$-2 + i$
b_i	$10 + 20i$	$5 + 10i$	$30 + 10i$	$10 + 20i$	25
$(f_i(A), f_i(B), f_i(C))$	(D, A, C)	(M_1, M_3, A)	(M_2, B, M_3)	(D, M_2, M_1)	(M_3, M_1, M_2)

Then f_i is a similarity with ratio $1/\sqrt{5}$ and $f_i(T) = P_i$. In particular the P_i are pairwise congruent and similar to T .

Proof. $|f_i(z) - f_i(w)| = |a_i| |\bar{z} - \bar{w}|/5 = \sqrt{5} |z - w|/5 = |z - w|/\sqrt{5}$, since $|a_i|^2 = 5$. The vertex images in the last row are direct computations. For example, $f_1(10) = ((-1 - 2i) \cdot 10 + 10 + 20i)/5 = 0 = A$ and $f_1(5i) = ((-1 - 2i)(-5i) + 10 + 20i)/5 = (-10 + 5i + 10 + 20i)/5 = 5i = C$. An affine bijection maps the convex hull of three points onto the convex hull of their images, so $f_i(T) = P_i$. Two similarities of the same ratio differ by an isometry, so the P_i are pairwise congruent. $\square$

A nondegenerate triangle is the intersection of the three closed half-planes that are bounded by its edge lines and contain the opposite vertex. Its interior is the intersection of the corresponding open half-planes. For T and the pieces this gives:

triangle	closed region (interior: all inequalities strict)
T	$x \geq 0, \quad y \geq 0, \quad x + 2y \leq 10$
$P_1 = ADC$	$x \geq 0, \quad y \geq 2x, \quad x + 2y \leq 10$
$P_2 = AM_3M_1$	$y \geq 0, \quad y \leq 2x, \quad x + 2y \leq 5$
$P_3 = M_3BM_2$	$y \geq 0, \quad 2x - y \geq 10, \quad x + 2y \leq 10$
$P_4 = M_1M_2D$	$y \geq 2, \quad y \leq 2x, \quad x + 2y \leq 10$
$P_5 = M_3M_2M_1$	$x + 2y \geq 5, \quad y \leq 2, \quad 2x - y \leq 10$

Each line passes through the two vertices of the corresponding edge, and the opposite vertex satisfies the inequality strictly. For instance, for P_5 : $x + 2y = 5$ passes through M_3, M_1 and gives $10 > 5$ at M_2 ; $y = 2$ passes through M_1, M_2 and gives $0 < 2$ at M_3 ; $2x - y = 10$ passes through M_3, M_2 and gives $0 < 10$ at M_1 .

Theorem 3. T is a rep-5 tile. Hence (1) is false: T is convex, and 5 is neither a perfect square nor 2.

Proof. By Lemma 2, the pieces $P_i = f_i(T)$ are similar to T and pairwise congruent. It remains to check conditions 3 and 4 of Definition 1.

$P_i \subseteq T$. Each vertex of each P_i is one of $A, B, C, D, M_1, M_2, M_3$, and each of these satisfies $x \geq 0, y \geq 0, x + 2y \leq 10$. Since T is convex, $P_i \subseteq T$.

$T \subseteq \bigcup P_i$. Let $(x, y) \in T$, so $x \geq 0, y \geq 0$ and $x + 2y \leq 10$.

- If $y \geq 2x$, then $(x, y) \in P_1$.
- Otherwise $y \leq 2x$. If moreover $x + 2y \leq 5$, then $(x, y) \in P_2$.
- Otherwise $y \leq 2x$ and $x + 2y \geq 5$. If $y \geq 2$, then $(x, y) \in P_4$.

- Otherwise $y \leq 2$ as well. If $2x - y \geq 10$, then $(x, y) \in P_3$, using $y \geq 0$ and $x + 2y \leq 10$.
- Otherwise $x + 2y \geq 5$, $y \leq 2$ and $2x - y \leq 10$, so $(x, y) \in P_5$.

Interiors are disjoint. For each pair $i < j$ we give an affine function ℓ with $\ell \geq 0$ on P_i and $\ell \leq 0$ on P_j . Then the interiors lie in $\{\ell > 0\}$ and $\{\ell < 0\}$ respectively. On a triangle, an affine function is nonnegative iff it is nonnegative at the three vertices, so this is a finite check.

pairs	ℓ	reason
$(1, 2), (1, 3), (1, 4), (1, 5)$	$y - 2x$	$\ell \geq 0$ on P_1 ; $\ell \leq 0$ at every vertex of $P_2, \dots, P_5$
$(2, 3), (2, 4), (2, 5)$	$5 - x - 2y$	$\ell \geq 0$ on P_2 ; $\ell \leq 0$ at every vertex of P_3, P_4, P_5
$(3, 4)$	$2 - y$	the vertices of P_3 have $y \in \{0, 2\}$; P_4 has $y \geq 2$
$(3, 5)$	$2x - y - 10$	≥ 0 on P_3 and ≤ 0 on P_5 (table above)
$(4, 5)$	$y - 2$	≥ 0 on P_4 and ≤ 0 on P_5 (table above)

The vertex values used are as follows. $2x - y$ equals 0 at A, D, M_1 , 10 at M_2, M_3 , and 20 at B . $x + 2y$ equals 5 at M_1, M_3 and 10 at B, D, M_2 . As a consistency check, the doubled areas are 50 for T and 10 for each P_i . $\square$

Remark 4. In general, a triangle is a rep- k tile iff $k = n^2$, $k = n^2 + m^2$ or $k = 3n^2$ [2]. The present example is the case $5 = 1^2 + 2^2$. Readers who know this theorem can stop here; the point of this submission is a complete, self-contained and machine-checked proof of one instance. The rectangle with sides 1 and $\sqrt{k}$ is a rep- k tile for every k (Section 4). So under reading (R1) the clause fails for every non-square $k \geq 3$.

4 The least-order reading: the $1 \times \sqrt{3}$ rectangle

Proposition 5. *Let $R = [0, \sqrt{3}] \times [0, 1]$. Then:*

- R is a rep-3 tile, with three congruent pieces that are rotated (orientation-preserving) copies of R ;*
- R cannot be dissected into 2 pieces similar to R , whether or not the pieces are congruent.*

Hence the least $k \geq 2$ for which R is a rep- k tile is 3, under both the congruent and the merely-similar notion. Since 3 is neither a square nor 2, the clause also fails under reading (R3), in all its variants.

Proof. (a) For $j = 1, 2, 3$ let $g_j(z) = iz/\sqrt{3} + j/\sqrt{3}$. This is a rotation by 90° composed with scaling by $1/\sqrt{3}$ and a translation. It maps (x, y) to $((j - y)/\sqrt{3}, x/\sqrt{3})$, so $g_j(R) = [(j - 1)/\sqrt{3}, j/\sqrt{3}] \times [0, 1]$. These three strips cover R , and their interiors are disjoint.

(b) Suppose $R = Q_1 \cup Q_2$, where $Q_i = h_i(R)$ for similarities h_i of ratio r_i , and the interiors are disjoint. Comparing areas gives $r_1^2 + r_2^2 = 1$, so $0 < r_i < 1$. Each Q_i is a rectangle with sides $r_i\sqrt{3}$ and r_i and diagonal $2r_i < 2$.

Every corner c of R lies in some Q_i . Since c is an extreme point of R and $Q_i \subseteq R$, c is an extreme point of Q_i , that is, a vertex of Q_i . A single Q_i cannot contain two opposite corners of R : they are at distance 2, which exceeds the diameter $2r_i$ of Q_i . So each Q_i contains at most two

corners, and the two are adjacent. The four corners are covered, so each Q_i contains exactly two adjacent corners, and the two pairs are complementary.

If Q_i contains the two endpoints of a side s of R , it contains s , by convexity. The line through s supports R , and hence Q_i . So Q_i meets that line in an edge of Q_i that contains s and lies in $R \cap (\text{line}) = s$. Thus s is an edge of Q_i . A long side ($\sqrt{3}$) is impossible, since $r_i\sqrt{3} < \sqrt{3}$ and $r_i < 1 < \sqrt{3}$. So Q_1 and Q_2 each have one of the two short sides of R , of length 1, as an edge. Then $r_i\sqrt{3} = 1$ or $r_i = 1$, and since $r_i < 1$ this means $r_1 = r_2 = 1/\sqrt{3}$. Now $r_1^2 + r_2^2 = 2/3 \neq 1$, a contradiction. $\square$

5 Lean formalization

The project `lean4/` (Lean 4.19.0, core library only, no Mathlib) builds everything from scratch. All decisive objects are defined explicitly: points, convex polygons and their interiors, similarities, rep- k tiles, and the claim.

The plane. A point `(Pt)` is an integer triple (x, y, w) with $w > 0$. It stands for the rational point $(x/w, y/w)$, and every point of $\mathbb{Q}^2$ has this form. `Pt.same` is equality of rational points $(xw' = x'w, yw' = y'w)$. The squared distance is `sqNum/sqDen` with `sqNum(P, Q) = (x_P w_Q - x_Q w_P)^2 + (y_P w_Q - y_Q w_P)^2` and `sqDen(P, Q) = (w_P w_Q)^2`.

Polygons. `orient A B P` is the 3×3 determinant of the homogeneous coordinates. Its sign is the sign of $(B - A) \times (P - A)$, since all $w > 0$. `ConvexPolygon vs`: the list has at least 3 vertices, and every vertex that is not an endpoint of an edge lies strictly to the left of that edge (counterclockwise, strictly convex position). `InPoly` (closed region) and `InPolyInt` (interior) require `orient ≥ 0`, respectively `> 0`, for every edge. Both are invariant under `Pt.same` (`InPoly.congr`, `InPolyInt.congr`).

Similarities and rep-tiles. `IsSimilarity f n m` means $m \cdot \text{sqNum}(fP, fQ) \cdot \text{sqDen}(P, Q) = n \cdot \text{sqNum}(P, Q) \cdot \text{sqDen}(fP, fQ)$ for all P, Q , that is, $|f(P)f(Q)|^2 = \frac{n}{m}|PQ|^2$. Such an f is automatically a well-defined map of rational points (`IsSimilarity.respects`). `RepTile vs k` asks for a list of k maps f_i , each with `IsSimilarity f 1 k` (squared ratio $1/k$, so the pieces are similar to the tile and pairwise congruent). It further requires:

1. every f_i maps the region into itself;
2. every point of the region is $f_i(Q)$ for some i and some Q in the region;
3. for $i \neq j$, no point of $f_i(\text{int})$ equals a point of $f_j(\text{int})$.

For a similarity, the interior of $f_i(\text{region})$ is $f_i(\text{interior})$, so these are conditions 3 and 4 of Definition 1. Fixing the ratio at $1/\sqrt{k}$ only makes the definition stricter (Section 2).

The claim. `Claim := ∀ vs, ConvexPolygon vs → ∀ k, RepTile vs k → (∃ m, k = m · m) ∨ k = 2`. This is (1) under reading (R1), and the main theorem is

`theorem conjecture_00000007709_false : ~Claim`

Proof structure in Lean.

- `ConjAffine` is the map $z \mapsto (a\bar{z} + b)/d$ in homogeneous coordinates. `ConjAffine.sqNum_apply` and `ConjAffine.sqDen_apply` are polynomial identities showing that it scales squared distances by $|a|^2/d^2$; they give `ConjAffine.isSimilarity`.
- The maps `f1`–`f5` are those of Lemma 2. `f_similar` shows each has squared ratio $1/5$. `f_vertices` (by `decide`) checks the vertex images.
- `g1`–`g5` are the inverses $g_i(z) = a_i\bar{z} + c_i$. `back1`–`back5` show that $g_i(P_i) \subseteq T$ and that $f_i(g_i(P))$ is the point P . With `into` ($f_i(T) \subseteq P_i$), this gives $f_i(T) = P_i$.
- `pieces_sub`, `cover` and `disjoint` are the three parts of the proof of Theorem 3. They are linear-arithmetic goals in the homogeneous coordinates, closed by `omega`.
- `T_rep5` : `RepTile T 5` assembles these facts. `five_not_square` and $5 \neq 2$ then give the main theorem.
- Non-vacuity: `all_convex` shows that T and all five pieces satisfy `ConvexPolygon`. `interiors_nonempty` exhibits an interior point (the centroid) of T and of each piece, so the disjointness condition is not vacuous.

There is no `sorry`, no `native_decide` and no added axiom. `#print axioms` reports at most `propext`, `Classical.choice` and `Quot.sound`; `omega` uses `Classical.choice`.

From $\mathbb{Q}^2$ to $\mathbb{R}^2$. The Lean statement concerns rational points. The proof of Theorem 3 above is written directly in $\mathbb{R}^2$, and it is the same linear case analysis that Lean checks. The real statement also follows formally from the rational one, as follows. The maps f_i are given by the same formulas over $\mathbb{R}$, and the distance identity is a polynomial identity. The pieces are closed and their union is closed. Rational points are dense in T , which is convex with nonempty interior and has rational vertices. So covering the rational points of T means covering all of T . If two interiors met, their intersection would be a nonempty open set, which contains a rational point; this would contradict the rational statement. Containment $P_i \subseteq T$ follows from convexity, because the vertices are rational points of T .

What is only in this report. Proposition 5 (readings (R3) and (R4)) is proved here and checked numerically in `verify.py` (exactly, in $\mathbb{Q}(\sqrt{3})$). It is not formalized in Lean.

6 Reproduction

```
cd lean4 && lake build          # Lean 4.19.0, core only
cd .. && python3 verify.py      # Python 3 standard library only
pdflatex report.tex && pdflatex report.tex
```

`verify.py` uses a method different from the Lean case analysis, with exact rational arithmetic. It checks:

- that $f_i(A, B, C)$ are the vertices of P_i and that $|a_i|^2 = 5$;
- that all pieces have squared sides 5, 20, 25 (one fifth of 25, 100, 125);

- that each piece lies in T (its vertices lie in T);
- that each pair of pieces is separated by an edge line of one of them;
- that the areas add up.

Containment, disjoint interiors and equal total area together imply covering: the uncovered part of T is relatively open in T , so if it were nonempty it would have positive area. The script also tests 14641 rational grid points of T one by one. Finally, it verifies the rep-3 dissection of the $1 \times \sqrt{3}$ rectangle exactly in $\mathbb{Q}(\sqrt{3})$.

References

- [1] S. W. Golomb, Replicating figures in the plane, *Math. Gazette* 48 (1964), 403–412.
- [2] S. L. Snover, C. Waiveris, J. K. Williams, Rep-tiling for triangles, *Discrete Math.* 91 (1991), 193–200.